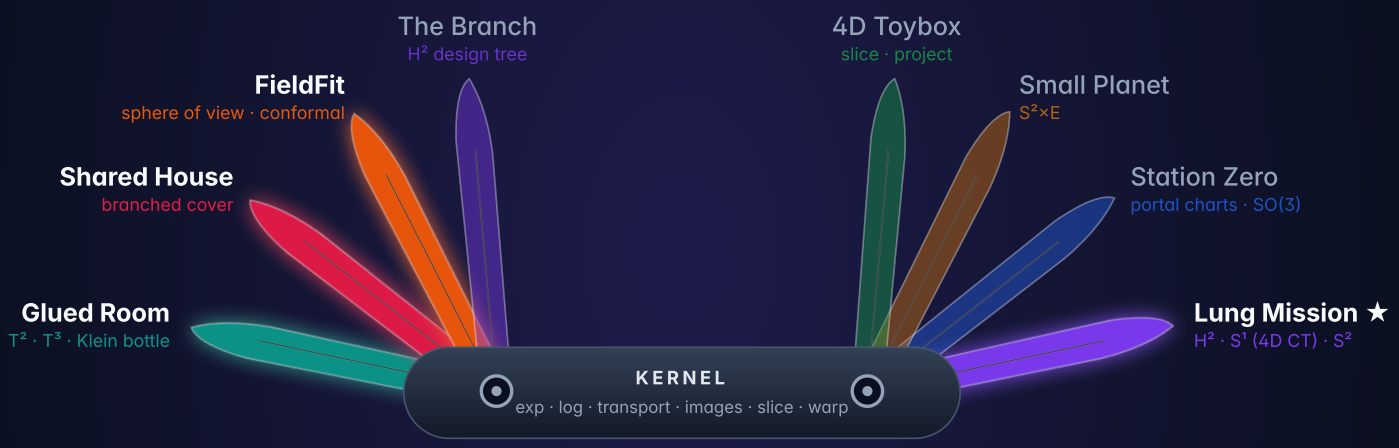
ATLAS

A Swiss Army knife for non-Euclidean interaction on a phone VR headset



Handle = one geometry engine. Blades = applications. Solid = in the live demo; faded = roadmap.

"Every space here has the shape of what's inside it."

WHAT IT IS

One phone headset that walks, sees your hands and knows your room, plus a design space of interactions that only non-Euclidean geometry makes possible, plus a set of **blades**: real applications where the space takes the shape of the problem inside it.

THIS DOCUMENT

The overall plan and what to use, for the team. The implementation agent gets the full brief `ATLAS_Implementation_Brief.md`, the engineering spec `ATLAS_Engineering_Spec.md` and the evidence `ATLAS_Research_Dossier.md` (≈850 links).

GEOMETRIES

E Euclidean · P portal · T glued · H hyperbolic · S spherical · B branched · 4D space / time · view warp

1	5	8	15	60	≤5
phone headset: iPhone 15 Pro Max	geometries in one live story	kinds of space, incl. 4D	hero interaction techniques	fps stereo, rendered on the phone	minutes in the headset per judge

ATLAS is a working title. In geometry, an atlas is a set of flat charts glued together, which is exactly how the engine works.

01 · THE IDEA

One headset, many blades

ATLAS is a platform, not a single app. One geometry engine (the handle) powers several short, purposeful applications (the blades). Each blade puts its content in the geometry it already has. Like a Swiss Army knife: you don't carve wood with it; you open cans, cut bandages, mark things.

The challenge, and how the plan scores

30%
Geometric creativity
The geometry *is* the content: trees in H^2, cycles on S^1, directions on S^2, homes on a branched cover. Live equations show it's correct.

30%
UX and interaction
Walk, look, press A. Hands and a marker cube are bonuses, each with a twin input. Blades are ordered for comfort.

25%
Graphics and tech
Locked 60 fps stereo on the phone; stencil portals, Klein-chart shader, 4D slicer, sky compute.

15%
Docs and demo
Knife Map dashboard, plan card, honest real-vs-pre-baked table.

Bonuses targeted: beyond gaming (medical training, assistive vision, remote care, science education) · educational value · polish in visuals and sound · novel techniques. The rubric's minimum is 30 fps; we target 60.

Three parts

1 · The handle (kernel)

One engine, five primitives: **exp/log map, parallel transport, gluing images, slice/project, conformal warp**, implemented for every geometry.

2 · The design space

Every interaction is a choice along six dimensions: **Geometry, Body, Mapping, Frame, Task, Perceiver**. 63 techniques catalogued; 15 heroes, 9–11 of them in the live path.

3 · The blades

Live story path: Bend Your Room → Breath Room → **Lung Mission** (airway, needle sky) → Post Circling, then FieldFit or House Call. **Roadmap:** Station Zero, The Branch, Small Planet, Manifold Lab, C-Space Room and more.

Four principles the evidence forces

1 Topology yes, global metric no

People handle portals and overlapping rooms well (small rooms can overlap by up to 56% unnoticed), but don't integrate a curved metric in their heads. So curvature creates room and adjacency, and people get flat local charts and landmarks.

2 Put curvature where the structure is

Branching (airways, trees)	H^2
Cycles (breathing, crystals, joints)	$S^1 \cdot T^2 \cdot T^3$
Directions (needles, beams, eyes)	S^2
Time series (4D CT)	4th axis
Many homes on one floor	branched cover

3 Pair every curved view with a flat twin

A curved view plus a linked conventional view is the only hyperbolic-interface configuration with a measured overall win (Risden et al. 2000). Curved gives overview; flat gives precision.

4 Measure performance, not applause

Users rate hyperbolic trees "cool" (7.7 vs 3.9) while performing no better. Every claim ATLAS makes has a live measure on the dashboard.

Non-negotiables

Safety never touches the network · every curved view has a flat twin · locked 60 fps (a deliberate, stable 30 fps thermal tier is the only allowed drop) · every hand technique has a twin input · a Euclidean **Safe Harbour** is one gesture, button or word away · redirection gains start at identity · no clinical claims · no camera frames stored.

02 · THE DEMO IN ONE WALK

One patient, five geometries, five minutes

The live path is one patient story told through five kinds of space, then one blade of the judge's choice. Before the headset goes on, the judge picks a **case card** (a public CT case) and hands over a pocket object, which a teammate photographs.

1. **Onboarding in passthrough** E 40 s
   Sees the real room through the camera, raises a hand, learns A and B.
   "That's every control: walk, look, press."

2. **Bend Your Room** H S 30 s
   Their own scanned room bends from saddle to flat to ball under a dial, floor still 1:1. They walk a taped square carrying an arrow, and it comes back turned by K·A.
   "Your feet just measured curvature."

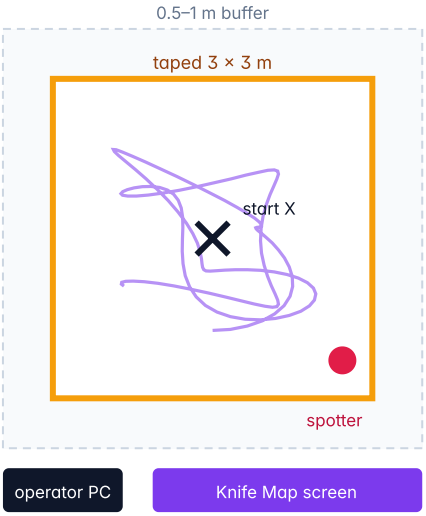
3. **Breath Room** flat, wrapped 4D time 45 s
   Walking east moves through the breath; the wall wraps 100% → 0%. The tumour is a tube through time; looking along it shows its shadow, the ITV. They flag the quiet part and see their own back in the next copy.
   "Time is a circle: flat, but it wraps."

4. **Airway** H^2 75 s
   The airway of the case they picked, completed procedurally (labelled), laid out in hyperbolic space. At one fork a straight-line compass and the hyperbolic beacon disagree; they choose. Their object waits as a cairn.
   "Straight-line intuition points the wrong way here."

5. **Needle sky** S^2 30 s
   Inside the lesion, every direction is a needle path coloured by risk; they pick one by looking. The plan card is generated.

6. **Post Circling** branched 20 s
   Through a door into a teammate's pre-scanned room; round one post and they're home again.

7. **Choice, then dashboard** FieldFit House Call 45 s + 30 s
   FieldFit or a live House Call; then results, decisions first: their compass choice, ITV saved, needle clearance, then path, comfort and frame time.



More than walking. Route planning, gating a beam, aiming a needle, handing over an object, measuring curvature with your feet, fitting a lens. Walking is the locomotion, not the point.

Comfort by design. Real walking only (no joystick motion). The world moves only while you hold a clutch. Topology before gains. ≤5 min in the headset. A Fast Motion Sickness score is asked before and after.

What onlookers see: the Knife Map screen beside the square

First-person mirror
The judge's view, with the live technique code as a chip, e.g. H · hand · exp · world · select.

Geometry × Body matrix
The live cell pulses; used cells fill in; empty cells are marked "open research".

Taxonomy ribbon
Six parallel axes; the current technique is a bright line over faded history.

Live equation card
The formula with real numbers, e.g. θ = |K|·A = 0.93 rad, measured 0.89.

Chart strip
The flat model of the judge's path: Poincaré disk, fundamental cell, sheet graph. It doubles as the flat twin.

03 · THE KNIFE

A design space of non-Euclidean interaction

Every technique is a choice along six dimensions, written as a code such as **H · hand · exp · world · select · self**. The empty cells suggest new techniques, which is the generative half of the HCI contribution.

G
Geometry: what space is the content in?

E · P portal · T glued · H hyperbolic · S spherical · B branched · 4s space · 4t time

B
Body: what drives it?

feet · head · hand · two hands · tangible (cube, scan, furniture) · voice · gamepad · partner · AI agent

M
Mapping: body motion → space motion

iso · gain · exp/log · holonomic · wrap · sheet · slice · conformal · jump

F
Frame: where is the geometry?

world (inside it) · window (portal, lens) · proxy (miniature, cube) · field (warped view) · twin (flat chart)

T
Task: what is the person doing?

navigate · select · manipulate · measure · create · communicate · learn

P
Perceiver: who notices?

self (overt) · covert · partner · audience

The handle: five primitives, every geometry

exp / log	transport	images	slice / project	warp
reach, step, ray, distance	carry frames; loops turn them	copies in glued and branched spaces	4D space and 4D time	conformal lenses on the view

One interface (`IGeometry`) with implementations for Euclidean, hyperbolic, spherical, glued, branched, portal and 4D. Write a technique once and it works in every blade.

The 15 hero techniques (each learnable in ≤30 s, each with a twin input, each producing a live number)

X1 Geodesic Ray
Point; the laser obeys the space

X5 Curvature Dial
Saddle → flat → ball, live

H1 Polar Tree Reach
Each hand-width = a generation

H2 Grab-the-Plane
Pinch the floor, drag the universe

H4 Foucault Cube
Walk a loop; the arrow turns by K·A

S1 Look-to-Aim
Every gaze direction is a needle

Q1 Breathing Ring
Walk a circle through a breath

D2 Slice Cube
Turn the cube to turn the 4D knife

S5 Blind-Spot Finder
Map your own blind spot

S3 Pinch-the-Sky
Pinch the world like a photo

B1 Post Circling
Round one post = another home

B3 Hand-over Through the Cut
Give Grandma the cup

T2 Backdoor Reach
Closer through the wall behind you

T3 Glue Die
Twist the cube to re-glue a wall

P1 Rotating Hatch
"Up" changes; point to the lifeboat

Plus **X9 Safe Harbour**, always on and taught first: two palms down, double-tap B, or say "home".

Three exact results that tie it together (slide-worthy)

Pinch-the-Sky is a Möbius map

A two-hand pinch fixes $w = a \cdot z + b$ on the sphere of view. The same group of maps ($PSL(2,\mathbb{C})$) flies you through H^3 and describes relativistic aberration: FieldFit's best 54°→10° lens equals the view at **0.94c**.

The cube is a 4D slicer

Its orientation is a unit quaternion, i.e. a hyperplane normal in $\mathbb{R}^4$. Turning it by θ tilts the slice by θ/2. On 4D CT the tumour's breathing becomes a tube whose shadow is the clinical **ITV**.

Gauss–Bonnet as a ruler

Walk any loop of area A; a carried arrow turns by $\theta = |K| \cdot A$. At R = 1.5 m, a 1.5 m square gives 1 rad. Judges **measure curvature with their feet**.

04 · THE BLADES

Each blade gives the space a job

Tier A is in the live demo and judges try it. Tier B is built if time allows and shown opt-in or on video. Tier C is the long-project roadmap.

BLADE · GEOMETRY	WHAT YOU DO (BEYOND WALKING)	WHO IT HELPS · EVIDENCE ANCHOR
Hub E your scanned room	Calibrate, return to safety. **Bend Your Room:** a dial bends the scanned room saddle → flat → ball at 1:1; a carried arrow measures K·A.	Everyone. The Euclidean "home chart" of the atlas.
Glued Room T^2 · T^3 · Klein · orbifolds	Walk through a wall and re-enter from the opposite one; reach through the "back door"; re-glue a wall with the cube; carry a mug round a flip. **Breath Room** (live path): the same engine with one axis as time, so the tumour becomes a tube whose shadow is the ITV.	Chemistry, robotics, AI-literacy and cosmology students. Periodic boundaries are how molecular-dynamics codes model crystals.
Lung Mission ★ H^2 S^1 S^2 deep blade	Pick a case card: **Roll a Lung** completes the airway procedurally (labelled) and lays it out in H^2. Walk the forks; at the **Two Compasses** fork choose between a straight-line compass and the hyperbolic beacon; pick a needle direction by looking; get a plan card.	Bronchoscopy, IR and radiotherapy trainees; educators; patient education. With 2D CT, path choice was right in 14–40% of cases vs 49–96% with virtual bronchoscopy.
Shared House branched cover + House Call	Live path: **Post Circling** on pre-scanned rooms (solo, no network). Option: a live House Call, where a teammate visits your scanned home through a doorway, pins hazards and times a chair-stand test.	Isolated older adults, families, occupational therapists. Home-hazard reduction cuts falls by 26%; flat telepresence placement succeeds in only 13% of 4-person sessions.
FieldFit sphere of view · conformal + Scotoma River	Map your blind spot; search under a real glaucoma field; fit a lens and find everything; read text that flows round a blind spot.	Low-vision researchers and rehab. Smartphone-VR perimetry r = 0.81 vs the clinical standard. Remapping exists (DSpecs); no optimal-within-class lens has been trialled.

Tier B · built if time allows

Station Zero P
Emergency egress where "up" changes at every hatch; point through the hull to the lifeboat.

The Branch / Deep Time H^2
Steer a generative design tree; walk from a photographed plant to its common ancestor with another organism.

Small Planet
$S^2 \times E$
Walk great circles, carry a Foucault pendulum, rerun an AI weather ensemble (incl. Pangu-Weather).

Manifold Lab S^1
Walk the circle an LLM uses for weekdays; your position steers what it says.

C-Space Room
T^2
The floor is a robot arm's joint space; a real arm follows you round obstacles.

Tier C roadmap: Cortex Planet (inflated brain) · Territory Surgeon (vessel tree) · Geodesic Soundscape (navigation by sound) · Wormhole Quest (citizen science) · Folded Building (a building in a small room) · Penrose Loop (Escher's ever-rising stair, honestly) · 4D Toybox · Embryo Room · Unit-Cell Room.

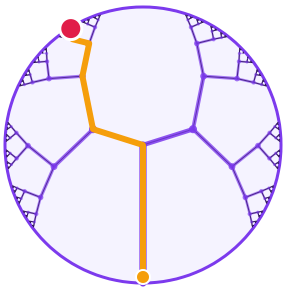
Where 4D earns its place. As **time**: the Breath Room, where the tumour is a tube through time whose shadow is the ITV (Tier B: the Heartbeat Torus, breath × heartbeat). As the w-coordinate behind overlapping rooms, and in a 4D toybox. Not as the core renderer.

Geometry strategy. Chosen: **portals + curved wings** (flat charts glued together; curvature only where the content is curved). Discussed: **true curved space only** (one continuous $H^2 \times E$ world): purer, but R ≈ 0.9 m in 3×3 m. A spike after Phase 2 decides.

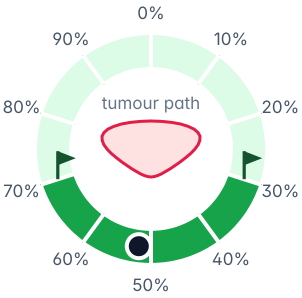
05 · THE DEEP BLADE

Lung Mission: one patient, three geometries

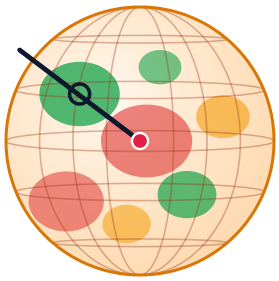
A public, de-identified 4D chest CT becomes a building with three wings. Each wing is in the geometry of its sub-problem, and each adds a line to the plan card. Framing: training and visualisation of public data, not for clinical use.



Airway wing · H^2
route to the lesion by walking forks



Breath Room · time circle
walk the breath; flag the quiet arc



Approach sky · S^2
every look is a needle; hold on green

PLAN CARD · TEACHING CASE

Route 4 forks · 112 mm · narrowest lumen 4.6 mm (scope 4.2 mm ✓)
Gating arc 30–70% · duty 40% · ITV 11.8 cm^3 (−32%)
Needle az 212°, el −8° · 18 mm of lung · 6 mm vessel clearance
Pointing probe closer to the Euclidean model
Values illustrative. Not for clinical use.

Roll a Lung: pick a case card; distal airways are completed by a published model (labelled) and laid out in H^2 in ≤10 s.

Why each geometry is essential, not decoration

Airway → hyperbolic plane

The airway is a near-binary tree of ≈23 generations. Hyperbolic circumference grows exponentially, so every branch keeps its width. At the **Two Compasses** fork a straight-line compass points the wrong way while the hyperbolic beacon points right. **Flat twin:** the true endoluminal view at every fork.

Breathing → circle (4D CT)

4D CT is sorted by breathing phase, so phase 90% sits next to 0%. A walked circle keeps that; a slider breaks it. Two flags bracket the quiet arc and the ITV and duty cycle update live. Tumours move 0–34 mm; motion over 5 mm must be managed (AAPM TG-76).

Needle → sphere of directions

Every straight path to the target is a point on a sphere, and so is every head direction: the body becomes the problem's coordinates. Risk is mapped on 49,152 equal-area HEALPix pixels (≈0.9°). Robust windows are those safe in all 10 phases.

Data pipeline (PC; whole collections on the H200)

Public CT TCIA 4D-Lung, LIDC-IDRI (CC BY 3.0) → **Segment** TotalSegmentator / MONAI: lobes, vessels, airways → **Skeleton** VMTK / skan graph + radii + labels → **Complete + lay out** distal airways by model → H^2 → tree.json → **Assets** meshes → GLB (meshopt, KTX2) + distance fields → **Phone** prefetched; renders offline

Done when: a naive user walks four forks (incl. the compass fork) in ≤75 s without help in ≥90% of runs · Roll a Lung builds a case in ≤10 s · the sky updates the current phase in <100 ms · the plan card prints with real numbers.

Honest limits: planners already auto-plan routes, so this is training and plan review. Distal airways are model-completed and labelled. The lung is in ordinary 3D space; H^2 holds its branching.

06 · THE SPACES, ILLUSTRATED

What kinds of space we're building

Every non-Euclidean space in ATLAS reaches your body in one of four ways. That is how one taped 3×3 m square can hold all of them.

1 · Copies and gluing

Every step is ordinary walking. Only crossing a wall, door or cut changes where you are. Exact and comfortable. *Glued Room, impossible rooms, Shared House.*

2 · Curved walking

Each footstep is carried into the curved space, one at a time. A flat floor can't be laid on a curved world without stretching, so where you end up depends on your path. *Hyperbolic wings, Small Planet.*

3 · Bent views

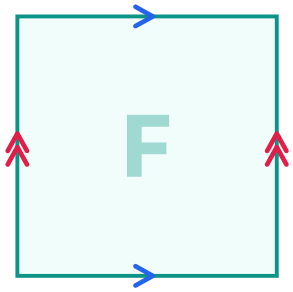
You stand still; the directions you see are remapped onto a sphere of view. *FieldFit.*

4 · Slicing

You stand in a 3D slice of a 4D object; your feet, the cube or time move the slice. *4D CT, 4D toybox.*

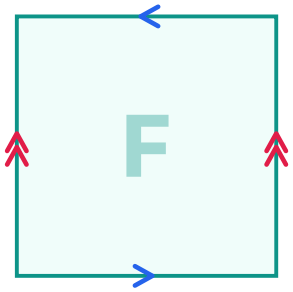
Glued Room: the flat torus and the Klein bottle $T^2 \cdot T^3 \cdot$ Klein

1 The room: a torus



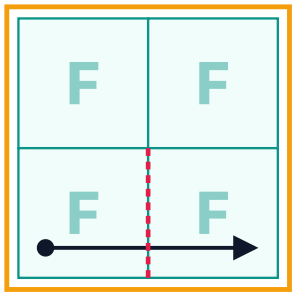
walls with matching arrows are glued

2 Variant: Klein bottle



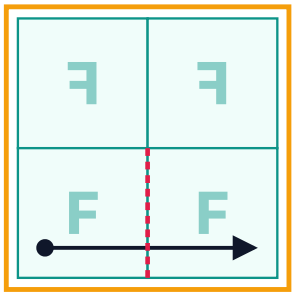
top ↔ bottom glued with a flip

3 On the real floor (torus)



3 × 3 m tape · 1.5 m room

4 On the real floor (Klein)



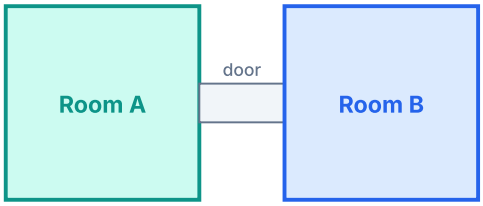
3 × 3 m tape · 1.5 m room

The virtual room is 1.5 m wide, so the taped floor holds 2×2 copies of it. Walk out of the east wall and you come in through the west wall of the same room; physically you just kept walking. **Your place in the room is your floor position "mod 1.5 m".** Look ahead and you see endless copies of the room, including the back of your own head. In the Klein version one pair of walls is glued with a flip, so every other copy is a mirror image, and a mug carried round the flip comes back mirrored. Adding the ceiling-to-floor gluing makes the 3-torus T^3: copies above and below too.

See it: Jeff Weeks' Torus Games · view from inside a 3-torus · Klein bottle as a glued square

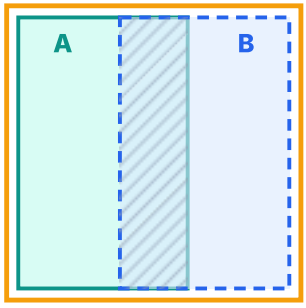
Impossible rooms: more building than floor portals

1 What you walk through



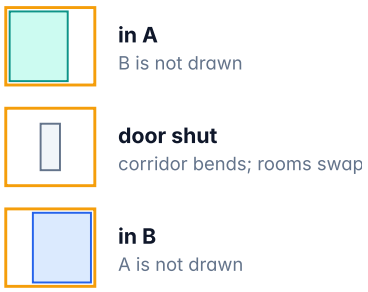
feels bigger than the room it is in

2 What the 3 × 3 m floor really holds



A and B share 40% of their floor

3 Never both at once



Two rooms reuse the same floor. You never see both at once: a door shuts or a corridor bends, and the next room is laid over the one you left. Small rooms can overlap by up to about half before people notice (Suma et al. 2012). This is how a building bigger than the room fits inside it, and it is the basis of the procedural landscape on page 10.

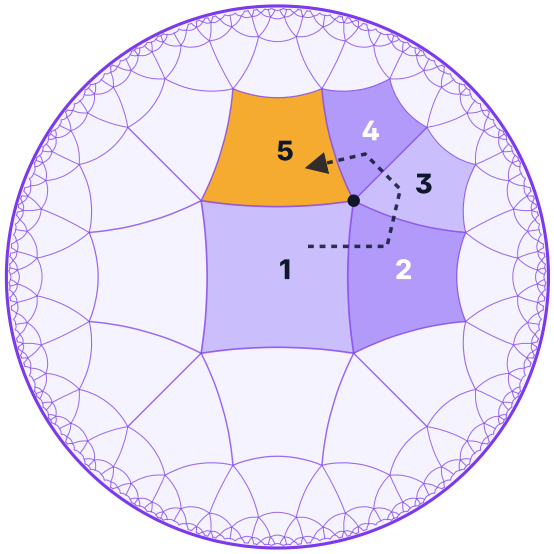
See it: Tea For God devlog · Unseen Diplomacy (video) · overlap diagrams (UMN)

Curved spaces: the world follows your path

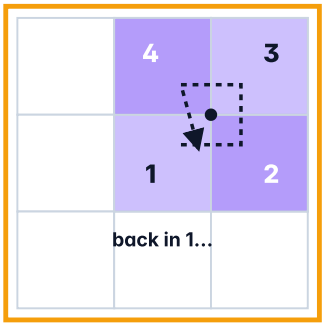
In a curved space we can't just copy the floor. Each step you take is replayed in the curved world, so two walks that end at the same spot on the floor can end in different places in the world.

Hyperbolic plane $H^2 \cdot H^2 \times E$

1 Hyperbolic floor: 5 squares per corner

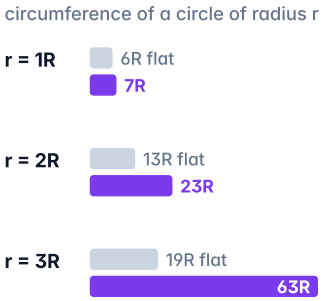


2 Real floor: 4 squares per corner



...but in the hyperbolic world you are in square 5: somewhere new. Loops never close.

3 More room further out



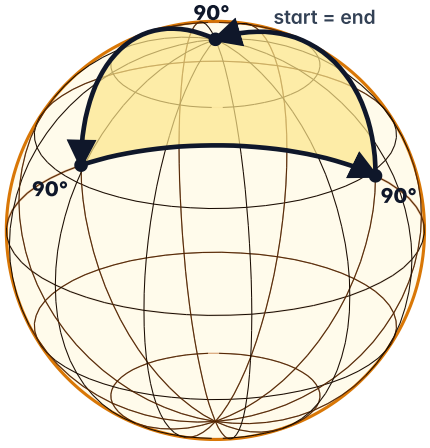
$2\pi R \cdot \sinh(r/R)$ grows exponentially: a tree that doubles every step fits without crowding. Airways do.

In the hyperbolic plane five squares meet at every corner, so there is more room the further out you go: a circle of radius 3R is over three times longer than on a flat floor. Map each 1 m square of the real floor to one hyperbolic square. Walk round a real corner (4 squares) and you pass only 4 of the 5 virtual squares, so you arrive somewhere new. Repeating loops gives an endless world in a 3×3 m room; this is the approach of Holonomy (TU Delft, 2024). At the radius this forces (R ≈ 0.9 m) it is strong and disorienting, so **Lung Mission uses fork chambers joined by short corridors, laid out in H^2 at R ≈ 2.2 m**. Walked spaces render as $H^2 \times E$: the floor is hyperbolic, "up" is still up.

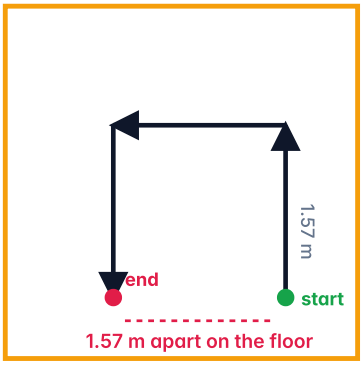
See it: the {4,5} tiling · HyperRogue · Hyperbolica · Holonomy paper

Sphere $S^2 \cdot S^2 \times E$

1 On a small planet: three right turns



2 The same footsteps on the real floor



3 Why they differ

Planet radius R = 1 m, so each leg is a quarter-circle: π/2 ≈ 1.57 m.

On the planet the triangle closes with three 90° corners (270°).

On the flat floor the same walk is an open square. The missing 90° is curvature × area: K·A = π/2.

A flat floor can't be laid on a curved world without stretching (Gauss), so the world follows your path, not your position.

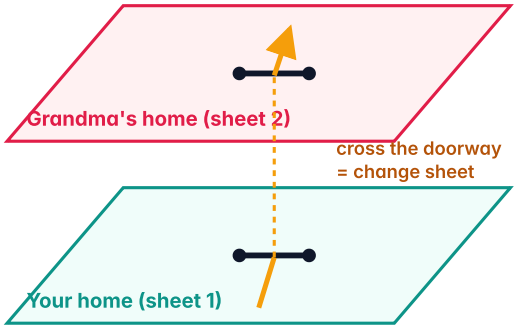
On a planet of radius 1 m, walk a quarter of the way round, turn 90°, and repeat twice: you're back where you started after three right angles. On the real floor the same footsteps make an open square that ends 1.57 m away. The missing 90° equals curvature × area (Gauss–Bonnet), and judges can measure it with the Foucault Cube.

See it: a triangle with three right angles · interactive spherical triangle · Curved Spaces

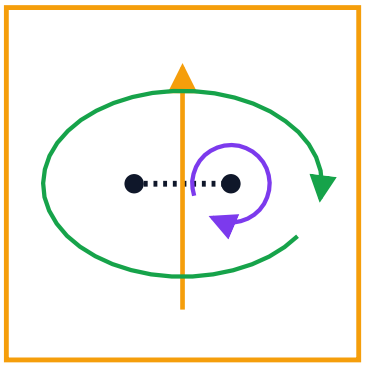
Sheets, views, and the whole set

Shared House: a walkable Riemann surface branched cover

1 Two homes stacked like car-park levels



2 In your real room: one free-standing door frame



through the door
→ Grandma's home

around the whole frame
→ still your home

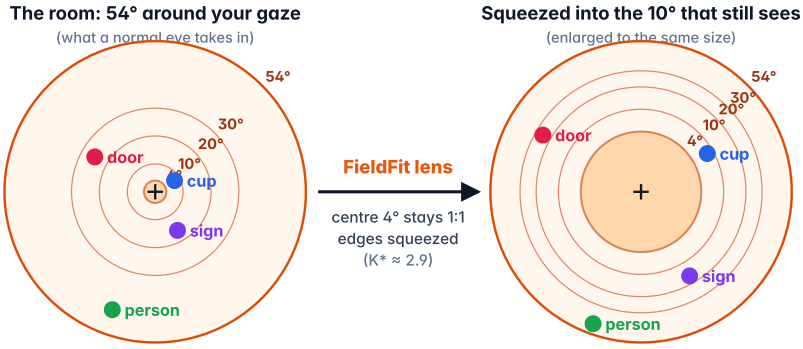
around one post only
→ Grandma's (crossed once)

All the curvature sits in the two posts (cone angle 4π); everywhere else is flat.

Each home is a "sheet". A free-standing door frame in your room is a cut between sheets. Walk through it and you're in the other home, at 1:1 in your own free space. Walk round the whole frame and you stay home; walk round one post and you switch. It is the Riemann surface of √z, made walkable, and the floor is flat everywhere except at the two posts.

See it: Riemann surface (spiral image) · the "infinite parking garage"

FieldFit: bending the view, not the floor sphere of view



Nothing moves underfoot; the camera image is remapped. Someone who sees only the central 10° gets the whole 54° room squeezed into it. The central 4° stays 1:1 so faces and text stay sharp, and the periphery is compressed smoothly. Lens modes are used standing or seated only.

See it: glaucoma vision simulator (AAO) · field-expansion devices (Peli lab)

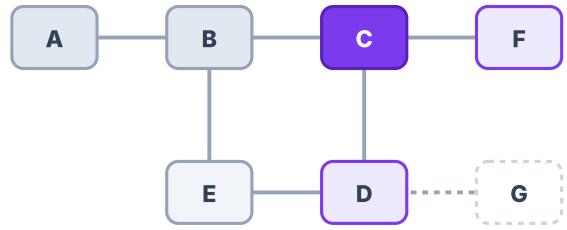
All the spaces at a glance

Space	Rule you learn in 10 s	How the floor maps	Where in ATLAS
T^2/T^3 · Klein	Every wall is a door to its opposite wall	copies (position mod 1.5 m)	Glued Room
portals	The building is bigger than the room	rooms re-placed out of sight	Hub, Station Zero, procedural worlds
H^2 · $H^2 \times E$	Five squares at every corner; branches keep their width	curved walking, step by step	Lung airway, The Branch
S^2 · $S^2 \times E$	Walk straight and come back	curved walking	Small Planet; the needle sky (a view)
branched	Through the door is another home	switch sheet at a cut	Shared House
view sphere	The room squeezed into what you can see	bent view (standing)	FieldFit
4D time · 4D space	You stand in a slice	slicing (feet, cube, time)	Breathing Ring, Slice Cube, 4D toybox

An endless world in a small room

A generated world has two layers: **the map** (which rooms exist and how they connect, permanent) and **the placement** (where each room sits on your real floor right now, recomputed whenever a room is out of sight).

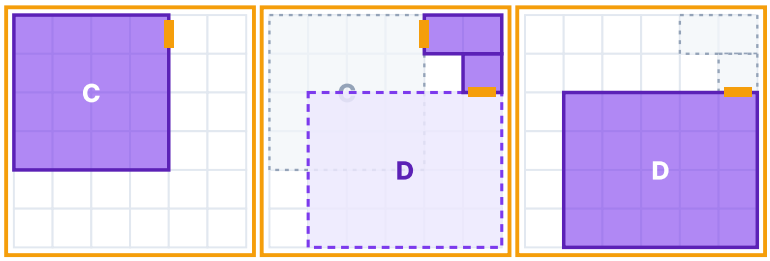
1 The world: a map of rooms, generated ahead



you are here
behind a door: placed next
visited (kept in memory)
beyond the fog: not made yet

loop B→C→D→E→B
seeded: same world
every time you return

2 The floor: each room is placed when you can't see it



in room C

in the corridor:
C's door shut, D placed

in D:
C's floor is free again

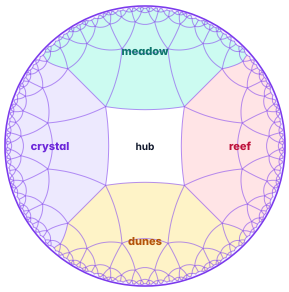
Rules: the floor is shrunk 0.3–0.5 m from the walls and cut into 0.5 m tiles. A room fits only if it sits on free tiles and each door (orange) opens onto a free tile. Rooms you can see together never overlap; rooms behind a closed door or a bend may overlap by up to ≈50%. Here C and D share 6 of C's 16 tiles.

Generated ahead, from a seed. Rooms are made a few doors ahead by a graph grammar that guarantees loops and keeps locked-door puzzles solvable. The same seed gives the same world when you come back.

Placed when you can't see it. The room behind a closed door is fitted onto free tiles of your floor, overlapping rooms you've left. Change blindness does the rest.

One look per region. A style cascade (region → room → props) picks palette, fog, materials and sound, so it reads as designed, not random.

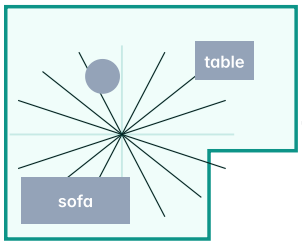
Data worlds aren't random. In Lung Mission the airway tree *is* the map; in Deep Time, the tree of life is.



Continuous curved landscapes. In a hyperbolic world each tile is generated when it comes within a few tiles of you, from its address (the moves from the hub), so it is identical every time. Branches never meet again in H^2, so whole branches become regions, like HyperRogue's "lands". The number of tiles grows exponentially with distance: it never runs out.

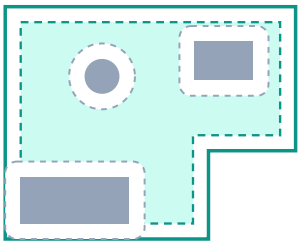
Mapping a real area

1 Scan the real space



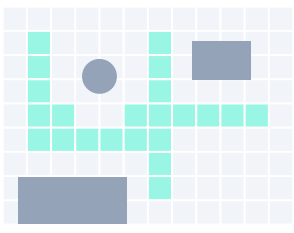
LiDAR mesh + RoomPlan

2 Keep away from walls



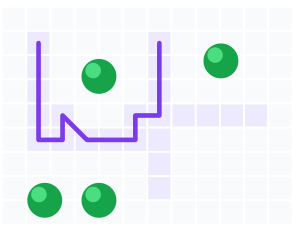
0.3–0.5 m safety margin

3 Cut into 0.5 m tiles



walkable tiles (teal)

4 Put the world on them



real obstacles → trees

In a bigger space (a hall, a gym, or outdoors in see-through mode) the phone keeps scanning as you walk. Only scanned, confirmed-free floor can hold virtual rooms; unscanned floor stays fogged. Real obstacles become in-world objects in the same place (trees, rocks, pillars), so you avoid them naturally. The 2 m / 1 m / 0.5 m collision warnings still run underneath, and a person walking in triggers passthrough.

07 · THE PROCEDURAL LANDSCAPE

Scanning and generation: which part does what

Two pipelines feed the worlds. **Scanning** brings your real room in and bends it. **Generation** builds worlds bigger than the room. They meet on the floor plan of the real room.

Scanning: the real room goes non-Euclidean live

Scan
RoomPlan inside the app: walls, doors, windows, furniture boxes

→

Rebuild
Modular kit: doors → portal sockets, furniture → walls and props

→

Tile
Free floor eroded into 0.5–1 m tiles; world map saved

Where it goes non-Euclidean	How
Bend Your Room	Your scanned room is placed in curved space vertex by vertex, then bent by the dial
Post Circling · House Call	Pre-scanned rooms become the sheets of a branched space
The judge's object	S24 photos → 3D model → a cairn in the hyperbolic airway
Real obstacles	Become a tree or pillar in the same place: the 2 m warning
Every generated room	Must fit the scanned free floor, one free tile past every door

Generation: worlds bigger than the room

Two layers. The map: rooms and connections grown from a seed, with loops; the same seed gives the same world. **The placement:** where each room sits on your floor now, re-solved whenever a door closes, so hidden rooms can share the same floor.

Generated world	Status
Roll a Lung: the patient's airway is the map; small airways grown by a published model (labelled), laid out in H^2, chambers placed on your floor tiles	live
Seed-generated worlds with a style cascade	roadmap
Endless H^2 landscapes, generated tile by tile from each tile's address	roadmap
Auto-dressed room interiors (wave function collapse)	roadmap

Escher-like spaces

Glued Room live

Every wall is a door to the opposite wall: copies of yourself in every direction, and the Klein-bottle version flips you into your mirror image.

Overlapping rooms live

Rooms that reuse the same floor out of sight: the building is bigger than the room. Plus the Breath Room's looping time and Post Circling.

Penrose Loop roadmap

Escher's ever-rising stair, built honestly: a spiral staircase whose top flight is glued to its bottom. Seated first, because a virtual ramp on a flat floor risks sickness.

Where each piece runs, and when

Step	Runs on	When	Time
Venue room scan	iPhone (RoomPlan)	day before + morning of	1–2 min per room
Kit rebuild and floor tiles	iPhone	after each scan	seconds
Bending the scanned room	iPhone (vertex shader)	live, every frame	within the 8 ms GPU budget
The judge's object	S24 photos → PC (SPAR3D)	during framing	≈20–60 s
Roll a Lung	PC	when the card is picked	≤10 s
Placing the next room	iPhone (tile solver)	whenever a door closes	one frame budget

> **In one line:** scanning makes *your* room the starting chart and the safety boundary; generation decides what lies beyond its doors.

08 · 4D, EXPLAINED

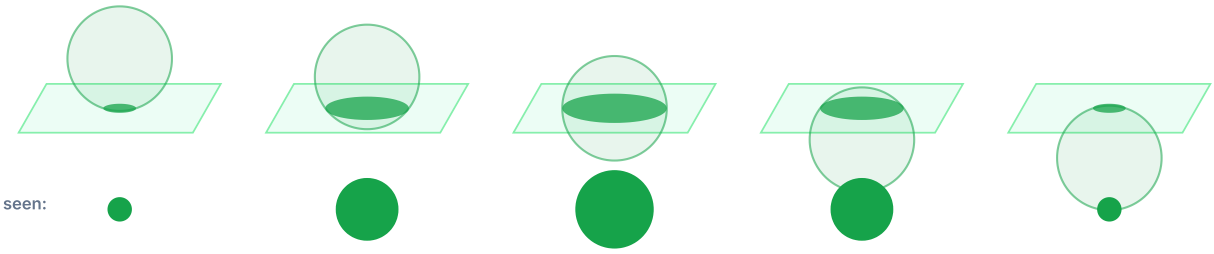
What "4D" means here

4D adds one more direction, called **w**, at right angles to the three we know. Nobody can see it directly, and ATLAS never asks you to. We always show a 3D **slice** or a 3D **shadow** of the 4D thing, and give you a handle (your feet, the cube, or time) to move the slice.

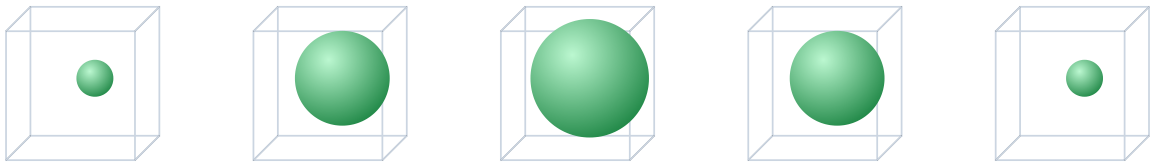
4D space (code 4s): shapes with four spatial directions, like the tesseract. Used in the 4D toybox and to explain impossible rooms.

4D time (code 4t): 3D space plus time as the fourth axis, like a 4D CT scan. Used in Lung Mission. The maths is the same: a 4D dataset, sliced.

1 A 3D ball passing through a 2D world: the flat creature sees a circle grow and shrink



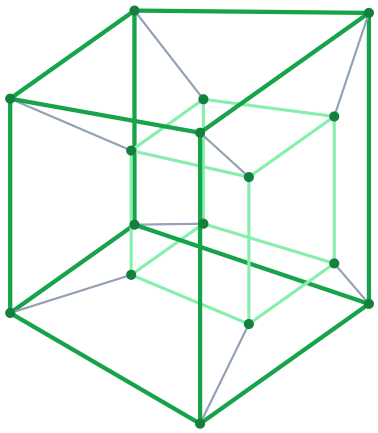
2 A 4D ball passing through our 3D world: we see a ball appear, grow, shrink, vanish



w = -0.95	w = -0.60	w = 0	w = +0.60	w = +0.95
radius $\sqrt{(1-w^2)}$ = 0.31	radius $\sqrt{(1-w^2)}$ = 0.80	radius $\sqrt{(1-w^2)}$ = 1.00	radius $\sqrt{(1-w^2)}$ = 0.80	radius $\sqrt{(1-w^2)}$ = 0.31

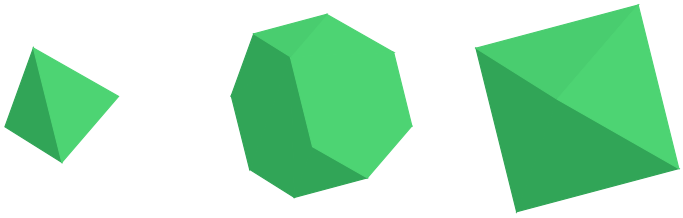
The classic way in (*Flatland*, 1884). A flat creature sees a ball passing through its plane as a circle that grows and shrinks. In the same way, we would see a 4D ball passing through our space as a ball that appears, grows, shrinks and vanishes. In the 4D toybox you see exactly this as you move w.

1 A tesseract (4D cube), shadowed into 3D



16 corners · 32 edges · 8 cubes
(the 'small cube' is the far cube, further in w)

2 Slicing it corner-first: the 3D cross-sections you would walk inside



near a corner	deeper	through the middle
tetrahedron	truncated tetrahedron	octahedron

Same object, sliced one step deeper each time. Move through a 4D object and what you see changes shape like this. The ATLAS slicer computes these cross-sections live on the phone's GPU.

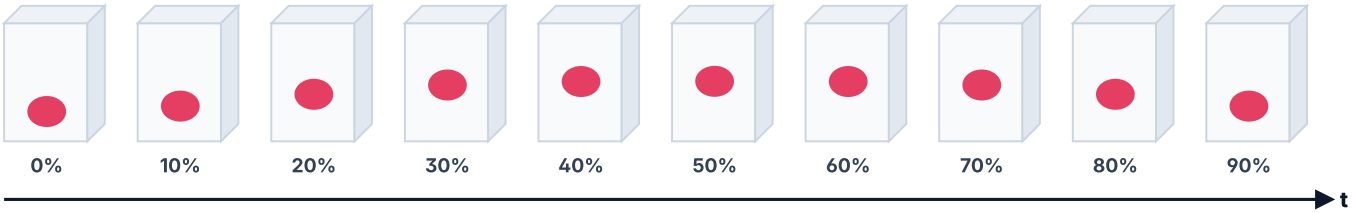
Left: a tesseract's 3D shadow, the familiar cube-in-a-cube. It is really 8 equal cubes, distorted by perspective the way a drawing of a cube distorts its squares. **Right:** its slices. Cut corner-first, the cross-sections run tetrahedron → truncated tetrahedron → octahedron (the middle) → and back again.

See it: rotating tesseract (Wikipedia) · Banchoff's 1978 hypercube film (projections and slices) · 4D Toys · Miegakure

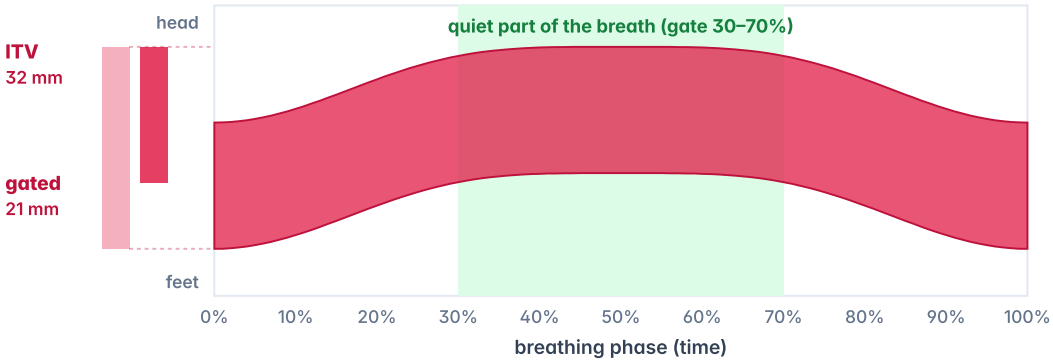
The 4D that matters: breathing

This is the real clinical use. A 4D CT scan records the chest through a whole breath. ATLAS lets you walk through that time and hold the slice in your hand.

1 4D CT = ten 3D scans, one per breathing phase



2 Stack them along time: the tumour becomes a tube in 4D (one space axis shown)

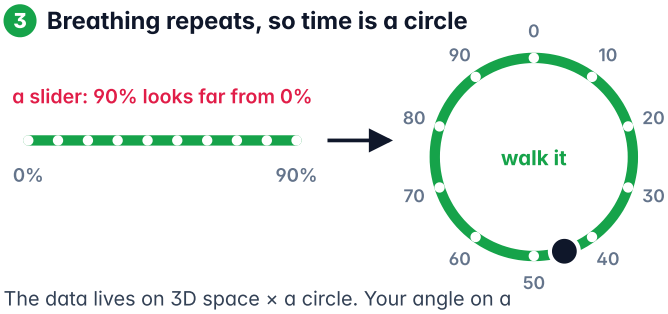


The ITV (internal target volume) is the tube's shadow along time: every place the tumour visits. Irradiate only in the green window and the shadow shrinks.

Values illustrative: 20 mm tumour, 12 mm motion (Lujan cos⁴ model).

A 4D CT is ten 3D CT scans, one per breathing phase. Stack them along time and the tumour traces a tube through 4D. The plot shows that tube with one space axis (head–feet) against time. The radiotherapy target that covers every position, the **ITV**, is the tube's shadow along time. If the beam is on only during the quiet part of the breath (gating), the shadow is shorter.

3 Breathing repeats, so time is a circle



The data lives on 3D space × a circle. Your angle on a 1 m floor ring is the phase; the chest breathes as you walk.

In the Breath Room, walking east moves you through the breath and the wall wraps 100% → 0% (the alternative layout is a 0.75 m floor ring). Flag the quiet part with two markers and the ITV and duty cycle update live.

Why this is the strongest 4D story. 4D CT is standard of care for lung radiotherapy: tumours move 0–34 mm with breathing, and motion over 5 mm must be managed (AAPM TG-76). Clinicians read it as ten separate 3D scans or a movie. ATLAS shows the motion as a *shape* you can walk round, and the ITV as that shape's shadow.

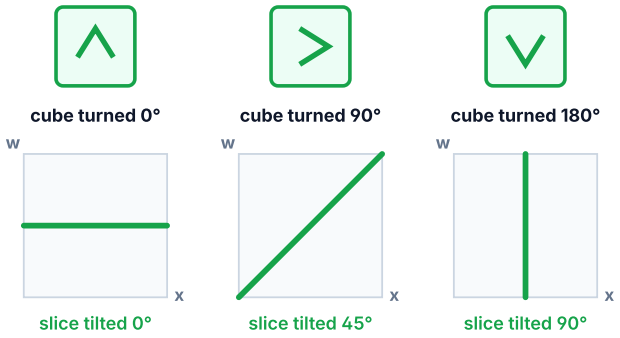
AAPM TG-76 report (Fig. 6: 4D CT sorting) · tumour volume across phases (open access)

Is the medical 4D non-Euclidean? Not by itself: 3D space plus a straight line of time is flat 4D, and slicing it is ordinary geometry. What makes it non-Euclidean in ATLAS is that breathing repeats, so time is a **circle**: the data lives on 3D space × a circle, which is flat but wraps around (the same kind as the Glued Room). The curvature in Lung Mission comes from its other two wings: the hyperbolic airway and the spherical needle sky. Say: *"time is a circle: flat, but it wraps."*

Holding the slice in your hand

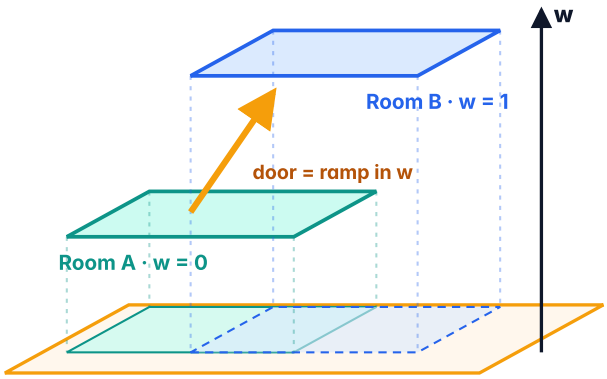
The printed marker cube is the handle for 4D. Its orientation is read by the phone's camera every frame and turned into the direction of the slice through 4D.

4 Turn the cube θ → the slice through 4D tilts θ/2



0°: ordinary x, y, z. 90°: halfway between x and w.
180°: x has been swapped for w (or for time, on 4D CT).
Why half? The cube's orientation is a unit quaternion
q = cos(θ/2) + sin(θ/2)·u, used as the slice's 4D normal.

5 Overlapping rooms don't collide in 4D



You always live in the slice at your current w. 'Peek' slides the slice so both rooms show through each other.

The Slice Cube on 4D CT. Hold the printed cube: its orientation sets which way the 3D slice cuts through 4D. A half-turn swaps one space axis for time. The tumour's motion then appears as a solid tube in front of you, and looking down its length shows the ITV.

Where 4D appears in ATLAS

Breath Room (Lung Mission)	4t live demo
Slice Tilt with the cube (Breath Room)	4t live, 10-min slot
Heartbeat Torus: breath × heartbeat	4t Tier B
"4D peek" through impossible rooms	4s demo moment
4D toybox: 4D ball, tesseract slices	4s roadmap
Tesseract House: 8 cube rooms, 3 per edge (S^3)	4s roadmap

What 4D is not, here: we never render a whole 4D world in 4D perspective. Every view is a 3D slice or shadow you can walk around.

How the slicer works (for the curious)

4D objects are stored as tetrahedra in $\mathbb{R}^4$ (the 4D version of a triangle mesh). Each frame, a compute shader cuts every tetrahedron with the slice hyperplane; each cut is a triangle or a quad, which the phone draws like any other 3D mesh.

Looking down past −35° fades to the 4D **shadow** on the floor instead (the projection along w, or along time on 4D CT, which is the ITV).

The cube's quaternion sign is kept continuous, so a full 360° turn of the cube tilts the slice 180°, and it takes 720° to come back to the start. That is real and teachable.

The challenge's 4D idea, made literal. "Just as a 3D object casts a 2D shadow, a four-dimensional object can be projected into our 3D world": the ITV is the 3D shadow of the tumour's 4D path, and Slice Tilt rotates the slice in a plane spanned by time and space, a rotation in a plane we cannot see. Production: 7 precomputed tilted slices per case (0–90°), blended; a live 4D slice is the stretch goal.

What the narrator says at this moment (Lung Mission, optional step): "Turn the cube half a turn. Now one direction in front of you is *time*. The red tube is the tumour across a whole breath. Walk round it: where it bulges, the tumour moves. Look straight down its length: that outline is the radiotherapy target." The spectator screen shows the slice angle and the ITV volume live.

09 · WHY EACH BLADE NEEDS ITS GEOMETRY

Where the non-Euclidean part earns its place

"Non-Euclidean" covers several different things. Saying exactly which kind each blade uses, and why its content has that shape, is what wins the 30% for geometric creativity and correctness.

curved
Constant curvature, not zero: hyperbolic (H^2, H^3) or spherical (S^2, S^3). Triangles don't add to 180°.

flat, wrapped
No bending, but the space closes up: torus, Klein bottle, a circle of time.

branched
Flat sheets joined at cuts; all the curvature sits at the cut's end points.

impossible gluing
Flat rooms joined so they overlap or can't fit in ordinary space.

directions
The sphere of directions around a point, and maps of it (lenses, needle paths).

4D
Flat 4D unless the 4D space itself is curved (S^3).

Argument strength: **STRONG** the content's structure forces this geometry · **MEDIUM** the geometry fits the structure and helps · **BASELINE** Euclidean on purpose.

Hub

Euclidean · BASELINE

WHY THIS SHAPE
Every curved world needs a flat reference. Your real room is the home chart and the Safe Harbour.

FLAT CAN'T...
Not applicable: it *is* the flat twin that every other blade is compared against.

HOW YOU MEET IT
Bend Your Room: the dial bends the room saddle → flat → ball at 1:1 underfoot, and a carried arrow comes back turned by K·A.

HONEST LIMIT
Only non-Euclidean while the dial is turned.

Glued Room

flat, wrapped · MEDIUM–STRONG

T^2, T^3, Klein bottle, orbifolds

WHY THIS SHAPE
Periodic things: a crystal is one cell repeated forever (simulations use exactly this "periodic box"); a robot joint wraps at 360°; a model's weekdays form a circle.

FLAT CAN'T...
In a closed box, a molecule just through the wall looks 1.4 m away instead of 0.1 m. No flat room can turn a mug into its mirror image by carrying it; a Klein bottle can.

HOW YOU MEET IT
The room repeats every 1.5 m: walk out through one wall and in through the opposite one.

HONEST LIMIT
Locally it is ordinary space; everything happens at the walls.

Lung Mission · airway

curved, negative · STRONG

H^2 (hyperbolic plane)

WHY THIS SHAPE
The airway splits about 23 times, doubling each time. Hyperbolic space has exponentially more room further out, so every branch keeps its full width.

FLAT CAN'T...
A flat plane runs out of room after a few generations; even a six-armed star has no greedy layout in the plane. In the real 3D lung, steering by direction misleads trainees.

HOW YOU MEET IT
Chambers and corridors laid out in H^2 ($R \approx 2.2$ m); walked distance grows with insertion depth; at the Two Compasses fork the straight-line compass and the beacon disagree.

HONEST LIMIT
The lung itself sits in ordinary 3D space; H^2 holds its branching. H^2 distorts 3D branch angles, so every fork also shows the camera view and a flat dendrogram.

Lung Mission · breathing

flat, wrapped · MEDIUM

time as a circle (S^1)

WHY THIS SHAPE
Breathing repeats. 4D CT is sorted by breathing phase, so 90% is next to 0%.

FLAT CAN'T...
A slider breaks the cycle at its ends; a walked circle keeps every phase next to its neighbours, at walking pace.

HOW YOU MEET IT
Breath Room: walking east moves through the breath; the tumour is a tube whose shadow is the ITV. Flag the quiet part and it shrinks. Slice Tilt turns the slice through time.

HONEST LIMIT
No curvature. 4D CT on its own is flat 4D; say "time is a circle: flat, but it wraps".

09 · WHY EACH BLADE NEEDS ITS GEOMETRY

Lung Mission · needle sky

directions · curved, positive · STRONG

S^2 of straight paths

WHY THIS SHAPE
All straight needle paths to one point are exactly the directions around it: a sphere. Where you look is also a point on that sphere.

FLAT CAN'T...
No flat map of a sphere is faithful (think world maps). Angle grids crowd near the poles and distort risk areas.

HOW YOU MEET IT
Stand inside the lesion; the sky is coloured by risk on 49,152 equal-area cells. Hold your gaze on green to choose. Two sweeps that start side by side meet at the pole; three windows make a triangle of more than 180°.

HONEST LIMIT
"Isn't it a skybox?" The answer is equal-area sampling, great-circle spacing, the great-circle check and windows safe across all 10 phases.

Shared House

branched · STRONG

walkable Riemann surface

WHY THIS SHAPE
Several real homes must each stay full-size, at 1:1, inside every visitor's own free floor.

FLAT CAN'T...
Flat telepresence shrinks to the overlap of everyone's rooms (placement worked in only 13% of 4-person sessions) or merges them into one.

HOW YOU MEET IT
Each home is a sheet; a door frame is the cut. Through it you visit; round one post you switch.

HONEST LIMIT
Closest prior art: MRUnion (2025). Phone-to-phone alignment is good to a few cm, so use snap zones.

FieldFit

directions · STRONG

conformal and quasiconformal maps

WHY THIS SHAPE
A visual field is a region of the sphere of directions; field loss removes part of it; a lens is a map between such regions.

FLAT CAN'T...
Shrinking a flat picture uniformly blurs the centre. The Schwarz lemma proves any shape-preserving lens must shrink the centre ≥5.8×, so ours bends shapes slightly at the edge instead.

HOW YOU MEET IT
The camera image is remapped so 54° of room lands in the 10° that still sees, with the centre kept 1:1.

HONEST LIMIT
Research prototype; display resolution limits acuity; standing only.

Tier B blades

Blade	Kind	Why this shape	What flat space can't do	Honest limit
Station Zero	impossible gluing	Each space-station module has its own "up": the station is local frames glued by rotations	No single flat map keeps every module's "up"; crews misjudge direction after hatches (Mir/ISS incidents)	Rotating "up" is the most sickness-prone; opt-in
The Branch / Deep Time	curved, negative	Design trees and the tree of life (2.39M tips) grow exponentially	Flat layouts crowd after a few levels	People rate hyperbolic trees "cool" without performing better; pair with a flat list
Small Planet	curved, positive	Weather lives on a sphere; a Foucault pendulum's daily turn depends on latitude, a holonomy of the curved Earth	Flat maps distort directions; globes beat flat maps for direction judgements	Small radius exaggerates curvature on purpose
Manifold Lab	flat, wrapped	Language models store weekdays and months on circles inside them	A straight axis cuts the circle: Sunday looks far from Monday	Topology, not curvature
C-Space Room	flat, wrapped	A joint angle wraps at 360°, so two joints make a torus	A flat square chart puts a seam at ±180°; short paths look long	Needs freely turning joints
Heartbeat Torus	flat, wrapped	Breathing and heartbeat are two independent cycles, so time is a torus	One slider or loop can't keep both cycles' neighbours together; east = breath, north = heartbeat does	A labelled composite (4D-Lung + ACDC heart MRI); topology, not curvature

09 · WHY EACH BLADE NEEDS ITS GEOMETRY

Tier C blades (roadmap)

Blade	Kind	Why this shape	What flat space can't do	Honest limit
Cortex Planet	curved, positive	The brain's surface is a folded sphere; neuroimaging already inflates it to a sphere for registration and "spin tests"	Flat brain maps need cuts	Mainly education
Territory Surgeon	curved, negative	Liver and lung vessel trees branch like airways	Same crowding as the airway	Segmentations stop short of fine vessels
Colon Gallery	flat, wrapped	The colon wall is a tube; conformal unfolding turns it into a corridor that wraps sideways	A fly-through alone sees ≈77% of the wall; the unfolded wall shows all of it	Hardest pipeline; show the distortion
Geodesic Soundscape	path distance	For blind travellers, sound should come from where the walkable path goes	Straight-line bearings point through walls	The floor is flat; only the distance is non-Euclidean
Wormhole Quest	impossible gluing	Impossible shortcuts separate map-based from graph-based navigation, relevant to dementia research	In flat worlds both strategies give the same answer	A research instrument, not a test or treatment
Folded Building	impossible gluing	Rehearse a real building's routes at 1:1 inside a small room	A 3×3 m room can't hold a building	It is impossible-space walking, and should be presented as that
House That Never Runs Out	curved, negative	Ancestors double every generation (2^n); shared ancestors become wormholes	A flat plan runs out of room after about 5 generations	Emotional, not clinical
Embryo Room	4D time, H^2	Development is 3D + time; cell lineages are trees	Flat lineage trees crowd	Time is linear here, so the 4D part is flat
4D Toybox / Tesseract House	4D, S^3	Slices of 4D shapes; 8 cube rooms, 3 around every edge	Eight cubes can't meet 3 per edge in flat 3D: they need curved S^3	The toybox is flat 4D; only the House is curved
Penrose Loop	impossible gluing	Escher's ever-rising stair is a spiral staircase whose top is glued to its bottom	No flat staircase keeps rising and returns to its start	A virtual ramp on a flat floor risks sickness: seated first

How to say it to judges

One sentence per blade, naming the kind.
"The airway is a tree, so it lives in hyperbolic space." · "Needle paths are directions, so they live on a sphere." · "Breathing repeats, so time is a circle: flat, but it wraps." · "Each home is a sheet; the curvature sits in the doorposts." · "A visual field is part of a sphere; the lens is a map of the sphere."

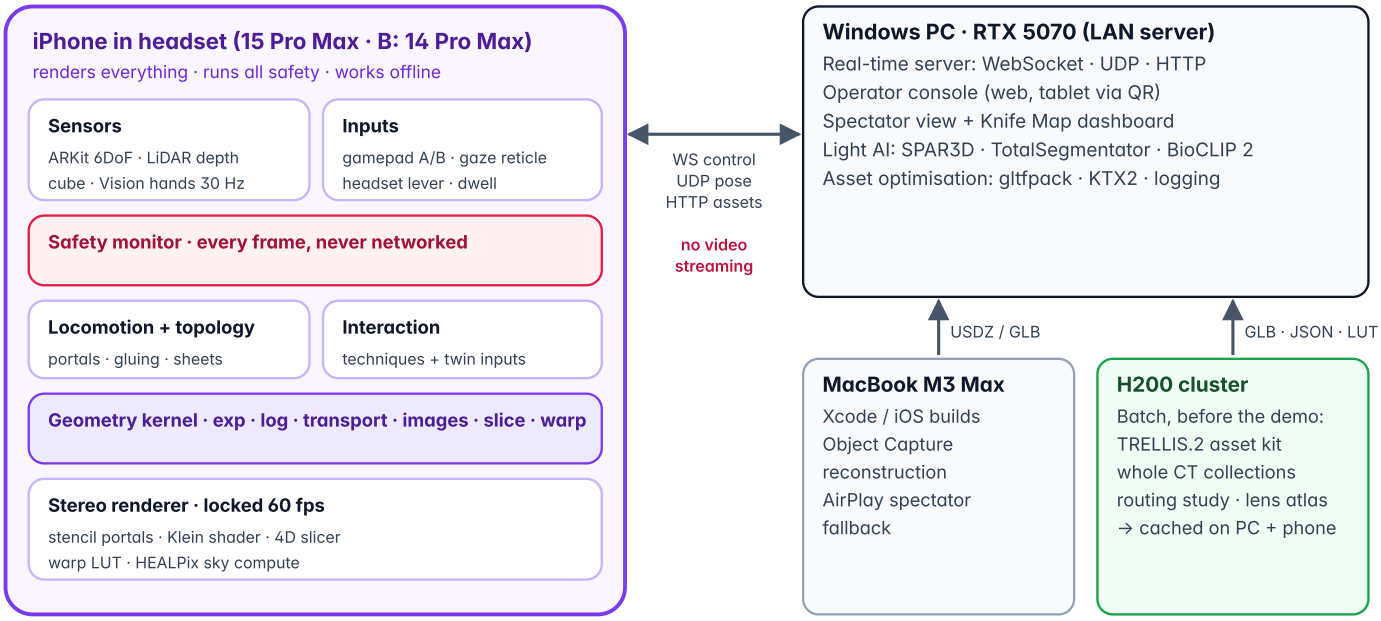
Don't over-claim. Don't say 4D CT is non-Euclidean (the wrapped time is). Don't call the Glued Room "curved" (it's flat and wrapped), or the lung "hyperbolic" (its branching is). Don't call a skybox a sphere without the equal-area maths. Every claim has a live number on the Knife Map.

The pattern behind all of them: curvature and gluing are chosen *from the content*: trees → hyperbolic, cycles → wrapped, directions → sphere, many homes → branched, time → slices. That is the rule on the cover: *every space here has the shape of what's inside it.*

10 · ARCHITECTURE

The phone does everything the user feels

Rendering and safety run on the phone, every frame. The PC suggests, logs and shows; the H200 cluster prepares assets in advance. Streaming VR video from a PC measures 50–150 ms against a ≈20 ms comfort budget, so we don't stream.



How the phone and PC pair

- Own 5/6 GHz router, no internet needed; PC wired to it.
- **WebSocket** for control, **UDP** for pose at 30–60 Hz, **HTTP** for assets by hash, **MessagePack** encoding.
- Clock sync at connect; every PC command expires in phone time.
- Development over USB: `iproxy` (iPhone), adb `reverse` (Android).
- iOS: Local Network permission; discovery by QR code or Bonjour.

When the network drops

- **Connected:** heartbeat within 250 ms.
- **Degraded** (250 ms–2 s): ignore stale plans, ease gains to identity, freeze world edits.
- **Offline** (>2 s): local planner and cached world; reconnect with backoff.
- **Resync:** clock and world-version delta.
- Every blade runs standalone by design.

Phone frame loop (order matters)

1. Poll inputs (pose, depth, cube, pad, hands)
2. Safety check and escalation
3. Locomotion: portal, glue and cut crossings
4. Interaction techniques → kernel
5. Blade simulation
6. Render each eye + distortion pass
7. Log (never blocks the frame)

Rendering in one paragraph. iOS has no single-pass stereo in Unity, so **one camera draws both eyes side by side** and a final pass does lens distortion, passthrough and a **late warp from a predicted pose**. Flat charts are joined by **stencil portals** (2.3× faster than render textures). Curved wings are stored on the hyperboloid or sphere and drawn through the **Klein chart** in the vertex shader, so straight edges stay straight. Walked spaces use $\mathbf{H^2 \times E}$ so the floor stays the floor. Geodesic fog; double-precision maths on the CPU.

Architecture principles. The world is a graph of reusable modules; the phone owns tracking, safety and rendering; commands are high-level; rooms reuse the same floor. The phone controller is primary, not a fallback. The PC only makes discrete choices behind closed doors ("which room next"), never continuous gains. Safety tables are precomputed per module.

11 · WHAT TO USE

Hardware and software

iPhone 15 Pro Max (phone A) + iPhone 14 Pro Max (phone B), Unity 6 with ARKit, a camera-window headset, a Python server on the RTX 5070 PC, and the H200 cluster for batch AI. The Galaxy S24 is the Android partner; Huawei is not required by the challenge document.

Phones: the line-up

iPhone 15 Pro Max phone A ✓	Main headset phone. LiDAR depth and mesh, RoomPlan, Object Capture, Vision hands. A17 Pro, 8 GB, USB-C.
iPhone 14 Pro Max phone B ✓	Has LiDAR too (A16, 6 GB): runs every feature on iOS 26/27. 10–26% slower under sustained GPU load; Lightning; 19 g heavier. Hot-swap and backup only, kept cool. **Lock both to its quality tier.**
Galaxy S24 / S24 Ultra	Photographs the judge's object; House Call partner client; Android test build. No depth sensor, so no on-device scanning and weaker collision warnings. Cardboard SDK listed S24 "black artifacts" in March 2026: test it.
Huawei	Not required by the challenge document (no ARCore; Unity 6 can't build for HarmonyOS NEXT).

Headset (buy two)

Must leave the **wide camera and LiDAR clear**, fit a 6.7" phone, have a head strap and vents. Should fit over glasses. Candidates: the **Zapbox shell** (shell only; its controllers can't run alongside ARKit) or a Merge-style AR headset; a cut Cardboard v2 for development. Generate a Cardboard viewer profile so lens distortion is exact.

Who does what

iPhones A and B	everything the user sees and does
M3 Max Mac	Xcode builds, Object Capture reconstruction
RTX 5070 PC	server, console, spectator, light AI (needs PyTorch cu128)
H200 cluster	TRELLIS.2 assets, CT collections, studies; cached
Galaxy S24	second participant, Android test

Software stack

Layer	Use
Engine	Unity 6 LTS · URP · IL2CPP · Metal; Burst/Jobs
AR tracking	AR Foundation 6 + ARKit (6DoF, depth, mesh, image tracking)
Stereo	One camera draws both eyes side by side (no single-pass stereo on iOS); Cardboard SDK for optics only; late warp from a Core Motion pose prediction
Native iOS	Swift plugins: Vision hands, RoomPlan, Object Capture, thermal state, Core Haptics
Assets	glTF + meshopt + KTX2 via glTFast; gltfpack on the PC
Server	Python FastAPI · asyncio UDP · msgpack; web console
Multi-user	LiveKit (voice), UniVRM avatars, Steam Audio
Scanning	Rooms: RoomPlan → rebuilt from a modular kit. Judge\'s object: S24 photos → SPAR3D on the 5070, live. Objects: Object Capture (fidelity) · SPAR3D or Hunyuan3D-2mini on the 5070 (speed) · TRELLIS.2 on the H200 (quality)
Medical	TotalSegmentator · VMTK · pylidc · HEALPix; data: LIDC-IDRI, 4D-Lung, UWHVF visual fields
Logging	UXF; replayable sessions
Study, don't copy	HyperEngine, SebLague/Portals, Hypermine, Tesserxel; HyperRogue is GPL, so read it only

Inputs

Must-have: real walking, head-gaze reticle, Bluetooth gamepad (A = use, B = back), operator console. **Showcase:** Vision hand pose lifted with LiDAR depth; a printed marker cube; real furniture as touchable props. **Avoid:** Zapbox controllers inside ARKit, ARKit audio capture (it mutes haptics), 120 fps.

Shopping list

No extra phone needed (the 14 Pro Max is phone B) · Lightning cable for it · 2 camera-window headsets that fit both camera bumps (+ Cardboard v2) · 2 Bluetooth gamepads · 2 printed marker cubes (8–10 cm) · Wi-Fi 6E router + Ethernet · 20–30 W USB-C chargers · desk fan · textured floor mat, gaffer tape, stanchions · disposable face covers, wipes, microfibre · A1 poster + safety sign · a free Apple account in Xcode (paid program only if needed).

12 · LIVE DEMO

A judge wears it

A 10-minute slot gives exactly 5:00 in the headset: the one-patient story, then the judge's choice. A 5-minute slot keeps the story (≈2:50, 3 forks, sky *or* Post Circling); a 3-minute slot shows Bend Your Room and the Breath Room live, the airway on video.

10-minute run-of-show (0:00–6:10; headset 0:40–5:40)

Frame	Onboard	Bend	Breath Room	Airway H²	Sky S²	Post	Choice	Exit	Results
0:00	0:40	1:20	1:50	2:35	3:50	4:20	4:40	5:25	5:40

During framing the judge picks a case card and a teammate photographs their object. 6:10–10:00: Q&A with the Geometry Lens hand demo, the classic Glued Room and Tier B videos, and the "Real / Pre-baked / Mocked" table. Answer "what's real?" first.

Roles

Narrator	talks to the judge, runs the script, asks the comfort questions
Spotter	**safety only, never merged with another role**; a shoulder touch means stop
Operator	PC console: blade queue, panic → passthrough, re-anchor, thermal tier
Explainer	at the Knife Map screen and the poster, for onlookers
Remote mate	House Call option only, with the S24 (a 4th person)

Collision safety (on the phone, every frame)

≈2 m	a virtual object appears where the real obstacle is
≈1 m	fading grid boundary
≈0.5 m	or a fast-approaching person, or tracking loss: automatic passthrough or full stop
always	audio + short haptic pulses; tape, buffer, spotter, the stop word "exit", ≤5 min

In public: two documented modes

Mode A · see-through recommended

Camera passthrough with overlays and 6DoF: continuous awareness of people and obstacles. In a 2024 field study, passthrough improved safety and social comfort in public without reducing presence.

Mode B · full VR with layered warnings

Only in a pre-scanned, roped-off zone with a spotter. Never walk in full VR in public without one. The ultra-wide lens mode pauses ARKit, so it is a seated research mode only; the demo uses Lens-6DoF.

If something fails

Tracking lost	fade to passthrough, re-anchor on the X; swap to phone B
Phone hot	drop to the stable 30 fps tier; swap phones between judges
Network / H200 down	blades are standalone; AI assets pre-baked and labelled
Partner absent	recorded avatar, labelled "recorded"
Hands fail	pad mode: every hero has a twin input
Total failure	3-minute sizzle video + earlier visitors' dashboard

Rehearsal targets: 15–20 dress rehearsals with naive people · median time in the headset ≤5:00 · Roll a Lung ≤10 s for every case card · ≥90% finish the Lung route without rescue · median exit sickness score ≤2 · 1%-low frame rate ≥45 fps on both phones after 30 minutes of back-to-back sessions · recover from a forced tracking loss in ≤15 s.

Onboarding (≤60 s)

Ask: "VR before? Carsick easily? Glasses?" (carsick → gentle path). Start in passthrough: strap, focus check, IPD nudge. "Raise a hand, touch the glowing button, press A; B goes back; say 'exit' any time." Fade to VR on the X.

Kit on the day

Both iPhones (same quality tier) and two headsets · own router, PC, Mac · chargers, fan · 2 cubes, 2 gamepads · face covers, wipes · poster (design-space map, "Real / Pre-baked / Mocked") · safety and consent sign (minimum age 13).

12 · LIVE DEMO

What we say, ask and sign

The narrator's script for the 10-minute slot, the exact questions, and the booth sign. Each space gets one rule and one "aha", so five geometries feel like one purpose.

Time	Narrator says	Cue
0:00	"Every space here has the shape of what's inside it. Today: one patient, five geometries. Pick a case card."	Operator starts Roll a Lung
0:15	"May we borrow something from your pocket? You'll meet it again inside the lung."	Teammate photographs it
0:40	"You'll see the real room first. Raise a hand… press A… B goes back. Double-tap B or say 'home' and you're back here."	Passthrough, then VR
1:20	"This is your own room. Turn the dial: the floor stays flat, but the space bends. Walk the square and carry the arrow."	Equation card
1:40	"It came back turned. That turn is the curvature times the area you walked: your feet just measured it."	—
1:50	"Breathing repeats, so here time is a circle: walk east and you move through the breath. Flat, but it wraps."	Breath Room
2:10	"Flag where the tumour is quietest. Look along the tube: that shadow is the radiotherapy target."	—
2:25	"Tilt the cube. You're rotating the slice through time: a turn in a direction you can't point to."	10-min slot only
2:35	"These are the airways of your card. They split about 23 times; hyperbolic space keeps every branch full width."	Airway
3:00	"Two compasses. The straight-line one points one way, the beacon another. Which do you trust?"	Log the choice
3:30	"Recognise this? Your object. Now point back to the trachea door and press A."	Pointing probe
3:50	"You're inside the lesion. Every straight needle path is a point on this sphere. Look at a green window to choose."	Needle sky
4:10	"Watch the two sweeps: they start side by side and meet at the top. On a sphere, parallels meet."	Great-circle check
4:20	"Through this door is someone else's home. Walk round the post… and you're back in yours."	Post Circling
5:40	"At the compass fork the straight-line compass pointed the wrong way; you chose [right/left]. About 10 m of airway inside 3 m of floor."	Dashboard reveal

Exact questions

Consent: "This is a research demo. We log your movement and answers under an anonymous ID; no camera images are saved. Say 'exit' at any time and we stop. OK?"
Comfort (before and after): "From 0, no sickness at all, to 20, frank sickness: where are you right now?"
Ease (after the airway): "How difficult or easy was finding your way? 1 very difficult, 7 very easy."
Spotter's briefing: "I'll stay right behind you. If I touch your shoulder, stop. Say 'exit' and we stop straight away."

Booth sign (A3)

ATLAS: walk through impossible geometry. A research and education demo on a phone VR headset, inside a taped 3×3 m square with a spotter; about 5 minutes.
Please don't try it if you are pregnant, have epilepsy or photosensitivity, a vestibular, balance or heart condition, or feel unwell. Minimum age 13.
We record movement and answers under an anonymous ID; **no camera images are saved**; ask us to delete your data any time.
Medical content uses public, de-identified scans for education only; not a medical device, not for clinical use.

Before each session (operator, 30 s): launch the app on the phone you'll use (a free-account build that won't open has expired: reinstall from the Mac over USB, ≈5 min, and use phone B meanwhile) · thermal state "nominal" or "fair" · re-anchor on the X · spotter in place · consent and comfort score ready on the console.

13 · BUILD PLAN

Spikes first, then one blade at a time

Phase 0 settles the platform unknowns with measurements on the slower phone (phone B) in the headset. Each later phase ends with something a naive person can try at 60 fps. The design space is *documented* in full; the story path and two choice blades are *built*.

Phase 0 ≈2 weeks · platform spikes	**Builds:** 24 benchmarks (below), run on phone B after a 20-min soak. Decides stereo (one camera vs two), stereo mode per wing, latency path; locks budgets.	**Done when:** every spike has a number and a decision; ≤1 missed vsync per minute.
Phase 1 kernel, hub, safety, tooling	**Builds:** kernel (E, T, P first), world model, one-camera stereo + final pass (distortion, passthrough, late warp), portals, hub from an in-app RoomPlan scan + kit, relocalisation, safety + Safe Harbour, pad/gaze input, Geodesic Ray, classic Glued Room, logging, message schema + server, console, Windows spectator, Knife Map v0, CI/CD, kill switches.	**Done when:** a naive user completes the Glued Room at 60 fps on phone B; Safe Harbour <300 ms; warnings at 2/1/0.5 m; commit-to-device ≤20 min.
Phase 2 story path core	**Builds:** Bend Your Room; Breath Room with Slice Tilt; Lung pipeline (case cards segmented on the H200, H² layout, tree.json, tile solver, chambers, valve doors); Roll a Lung; Two Compasses; endoluminal view; sphere-traced needle sky; plan card; pointing probe; object pipeline (S24 photos → SPAR3D → cairn); greedy study.	**Done when:** four forks in ≤75 s by a naive user; Roll a Lung ≤10 s; sky phase <100 ms; plan card validates.
Phase 3 Post Circling + choices	**Builds:** solo Post Circling on pre-scanned sheets (recorded avatar); FieldFit (blind spot, circular tunnel + power-law lens, Lens-6DoF); minimal S24 partner client + voice for the House Call option.	**Done when:** both choices run inside the 10-min show; recorded fallbacks exist.
Phase 4 hands, polish, show	**Builds:** Vision hands + twins; Geometry Lens; cube (Glue Die, 4D views); style, audio, earcons; dashboard reveal (decisions first); thermal tiers; videos; 15–20 rehearsals; docs.	**Done when:** all rehearsal targets in the live-demo section are met.
Phase 5+ long project	**Builds:** Tier B blades, full Android build, co-walking, Wormhole Quest, replications, studies with ethics approval, clinical partners. Spike 0.14 runs after Phase 2.	**Done when:** papers, studies, partners.

Phase 0 spikes

0.1 Stereo pipeline on both phones: GPU ≤8 ms at 150k tris/eye
0.2 ARKit through the window: FOV, LiDAR clear, drift <2%/20 m
0.3 Hand reach for 5 people; Geometry Lens detector ≥90%
0.4 LiDAR hand depth at 25/35/50 cm
0.5 Marker cube: jitter, latency, loss under rotation
0.6 Stencil portals in stereo: <1.5 ms at depth 2
0.7 Klein vs equidistant stereo per curved wing
0.8 Closed-headset thermal soak: no tier drop before 20 min
0.9 Wi-Fi round trip: p99 <30 ms on the LAN
0.10 Lens-mode switch (research mode): resume <2 s
0.11 RTX 5070 model timings; PyTorch cu128
0.12 H200 policy: batch only or interactive?
0.13 Cardboard viewer profile on both phones
0.15 One camera vs two: main + render ≤7 ms
0.16 Motion-to-photon with prediction + late warp: ≤30 ms
0.17 Vision on the Neural Engine: no GPU cost, no frame drops
0.18 RoomPlan inside the app's AR session; relocalise <3 cm
0.19 Airway depth, skeleton, H² layout, greedy success
0.20 Needle sky, 10 phases: <300 ms, ≤100 MB
0.21 3-device Wi-Fi: loss <1%, p99 <50 ms
0.22 Memory: all blades 20×, no jetsam, ≤1.8 GB
0.24 CI: commit to installed app ≤20 min
0.25 glTFast 5 MB GLB: worst frame <20 ms
0.26 Free Xcode signing on both phones; memory capability

Engineering standards: no per-frame allocations · double-precision CPU geometry · kernel identity, GPU-parity and golden-image tests · nightly perf replay on phone B · CI on the M3 Max · memory ≤1.8 GB · kill switches · licence ledger.

14 · PLANNED VS THE FULL IDEA

What gets built for the event, and what stays on paper

The full idea is a platform: 8 kinds of space, 63 techniques, about 20 applications. The build plan turns a deliberate slice of it into a live, judge-tested system; general procedural worlds, Tier B/C blades, H^3 and S^3 rooms, co-walking, full Android parity and the studies stay on paper.

● build-ready: inputs, outputs, approach, data formats and an acceptance test · ◐ specified: approach and parameters set, some interfaces still to write · ○ concept: described, not specified

Area	The full idea	Built for the event	Ready	Phase
Geometry kernel	8 kinds of space + lens maps, one interface	All 8 + lens maps; the live path uses 7	◐	1–4
Rendering	Stereo, portals, curved space, 4D, sky, lens at 60 fps	One-camera stereo, portals, Klein shader, slicer, sphere-traced sky, final pass with late warp; frame budget set	● portals · ◐ rest	0–2
Sensing and input	Walking, LiDAR, hands, cube, pad, gaze	All; every hand move has a twin; Geometry Lens as the hand hero	● pad, gaze · ◐ hands, cube	1, 4
Safety	Collision warnings, also in public	Two-source warnings with hysteresis; passthrough Safe Harbour; modes A/B	◐	1
Scanning → world	Rooms and objects scanned in, AI meshes	RoomPlan in the app → kit rebuild; the judge's object meshed live from S24 photos	◐	1–3
Procedural generation	Continuous non-Euclidean worlds	**Live:** Roll a Lung. General generated worlds: roadmap	◐ · ○	2 · 5+
Pathfinding	Routing and placement in curved and glued space	Airway route cost, Two Compasses, greedy study, needle clearance; general portal routing: roadmap	◐ · ○	2 · 5+
4D	4D space and 4D time	Breath Room (time circle, ITV as a shadow); toybox and Tesseract House: roadmap	◐ · ○	2 · 5+
Applications	≈20 blades	Story path + FieldFit + House Call + classic Glued Room; Tier B on video; Tier C in docs	◐ · ○	1–4
Design space	6 dimensions, 63 techniques	9–11 techniques live; all 63 documented; every technique logs its code	◐	1–4
Multi-user	Shared worlds, remote care	Solo Post Circling; live House Call as an option; co-walking: roadmap	◐ · ○	3 · 5+
PC side	Server, console, spectator, dashboard	One message schema, Windows spectator build, Knife Map live and reveal views	◐	1–4
AI and H200	Generation, segmentation, studies	Batch: case cards, tubes, sky volumes, greedy study, asset kit. Live: object mesh, Roll a Lung	◐	2–3
Engineering	Optimised and solid	Frame budget, tests, CI/CD, perf regression, memory budget, kill switches, crash reports	◐	0–1
Demo operations	A demo judges can try	5:00 run-of-show, roles, fallbacks, rehearsal targets	●	4

Schedule templates (pick one when the event date is known)

Weeks	Plan	Cuts built in
10 or more	W1–2 Phase 0 · W3–4 Phase 1 · W5–7 Phase 2 · W8 Phase 3 · W9 Phase 4 · W10 rehearsals	Only the standard cut list if needed
8	W1–2 Phase 0 · W3–4 Phase 1 · W5–6 Phase 2 · W7 Phase 3 (FieldFit only) + Phase 4 essentials · W8 rehearsals	Live House Call, Steam Audio; Slice Tilt as stretch
6	W1 make-or-break spikes · W2–3 Glued Room + Bend Your Room · W4–5 Breath Room, airway on curated cases, precomputed sky · W6 rehearsals	Post Circling and the judge's choice on video
Under 6	The 3-minute path (Bend Your Room + Breath Room + dashboard); the rest on video	Most blades

Gates: Phase 0 go/no-go per spike · Phase 1: Glued Room at 60 fps on phone B · mid-Phase 2: first full story run by someone new · T−7 days feature freeze · T−3 golden build · T−1 install and offline launch check. Owners per area: brief §10.8.

How the plan answers "Beyond Euclid"

The challenge asks for an application where non-Euclidean geometry is central and meaningful, so users "feel it, interact with it, and discover something new about space itself". It sets no hardware requirement and asks for 30+ FPS.

The challenge asks for	Where ATLAS answers it
Hyperbolic geometry	The patient's airway tree laid out in H^2, with exponential room so every branch keeps full width; the saddle end of the curvature dial
Spherical / elliptic geometry	The needle sky: every straight path is a point on a sphere; sweeps that start side by side meet; three windows make a triangle of more than 180°. The ball end of the dial; elliptic space as the one-pass S^3 option
4D projections	The Breath Room: the tumour's path through space and time is a 4D tube; its end-on shadow is the radiotherapy target; Slice Tilt rotates the slice in a time–space plane
Impossible spaces, Escher-like worlds	The Glued Room (every wall a door; Klein-bottle mirror flip), Post Circling (round one post into another home), rooms that overlap out of sight; the Penrose Loop on the roadmap
Coordinates, rendering, navigation rethought	Charts and gluings instead of global coordinates; the Klein chart in the vertex shader; exp-map walking; stencil portals; topology-first locomotion in 3×3 m
"Feel it, interact, discover"	Your feet measure curvature; straight-line intuition fails at the Two Compasses fork; looking chooses a needle path

Three moments that make the geometry felt, and correct

Slice Tilt · 4D

Tilt the cube and the Breath Room's slice rotates through time: the tube morphs into the tumour's 3D shape. A rotation in a plane you can't point to, in 10 seconds.

Great-circle check · S^2

Two needle sweeps leave the equator side by side and meet at the pole; the plan card shows the triangle's angle sum above 180°. Spherical geometry, not a skybox.

Say it exactly · H^2

The lung sits in ordinary 3D space; what lives in hyperbolic space is its branching structure. That is the claim the maths supports, and it answers the obvious judge question.

Against the judging criteria

30% **Geometric creativity**
Each geometry chosen because the data has that shape; every blade names its kind; live numbers.

30% **UX and interaction**
Walk, look, one button; twins for every gesture; Safe Harbour; one rule per space.

25% **Graphics and tech**
60 fps stereo target (30+ required); one-camera stereo; late warp.

15% **Docs and demo**
Brief, engineering spec, this plan, dashboard reveal, videos.

Main risk to the score: UX is 30%, and the challenge wants the experience "purposeful, not disorienting for its own sake". Five geometries in five minutes must not feel like a tour: one rule and one "aha" per space, tied together by the patient story and the plan card.

What could go wrong, and what to say

Top risks

Risk	Level	Mitigation
Thermal throttling over a demo day	high	Fixed 60 fps, thermal tiers, meshing off after mapping, vented headset, fan, second phone hot-swap
Scope creep (63 techniques, 15+ blades)	high	Tier A only until Phase 4 ends; the design space is documented, not all built
Hands not seen (narrow camera cone)	high	Hands are always optional; targets 20–30° below eye level at 40–55 cm; good lighting
Tracking loss (dark hall, plain floor)	medium	Textured mat, even light, saved world map, re-anchor on the X, passthrough on loss
Cybersickness	medium	Topology before gains, ≤5 min, comfort ordering, vignette, stop rule; rotating-"up" blade opt-in only
"It's just portals / a skybox"	medium	Live equations, Gauss–Bonnet check, walk-round-vs-through demo, one real task per blade
Over-claiming medically	medium	Claims wording below; public data only; "not for clinical use" on screen
Headset doesn't fit both phones or glasses	medium	Buy two candidates now; test with both iPhones' camera bumps and a glasses wearer
No single-pass stereo on iOS (CPU cost)	high	One camera draws both eyes; portals as extra draw lists; spike 0.15
Airway segmentation too shallow	high	Benchmark MONAI / nnU-Net; Roll a Lung completes distal airways (labelled); curated default case
Five geometries feel like a tour	medium	One rule and one "aha" per space; the patient story and plan card tie them together
Event details unknown	unknown	Ask the organisers now (questions below); the document sets no hardware requirement

Claims language (for the docs and the booth sign)

Say: "An interactive platform for exploring non-Euclidean interaction. Medical blades are educational visualisations of public, de-identified data with illustrative risk metrics. Not a medical device; not for clinical use." · "Research prototype that simulates visual-field loss and lets researchers prototype remapping lenses."

Avoid: "plans your procedure", "detects MCI", "rehabilitates", "expands your visual field", "treats…"; normative cut-offs; "first non-Euclidean VR" (Hart/Segerman, HyperRogue, Hyperbolica and Holonomy exist); "intuitive" without data; "AI-generated live" when it was pre-baked.

Basis: FDA General Wellness guidance (updated January 2026) and Health Canada SaMD guidance, where intended use decides device status. Human-participant studies need ethics review regardless.

Open questions to answer early

1. Event details: date and slot length? Code allowed before the event? Single judges or a panel? Venue space, lighting, Wi-Fi? (No hardware requirement in the challenge document.)

2. Team roles: the demo needs at least three people (narrator, spotter, operator).

3. Phones: confirm phone B is a 14 *Pro* Max (LiDAR), not a 14 Plus; test the S24 with the Cardboard optics.

4. H200 policy: batch only, or interactive nodes reachable from the venue?

5. Headsets: order two candidates now and test the fit.

6. Lung data: one 4D-Lung patient throughout (A) or a labelled composite (B)?

7. Blade set: keep Shared House, or swap in Station Zero if the second phone is a risk?

8–9. Name and geometry strategy: keep "ATLAS"? Add a true-curved-only mode after spike 0.14?

10–11. Staffing and case cards: a 4th person for live House Call? Which ≈20 public cases go in the deck?

17 · SHORT ANSWERS

The founding questions, answered

What is the purpose?

A platform for non-Euclidean interaction, with blades where the space takes the shape of a real problem: medical training (Lung Mission), assistive vision (FieldFit), social and remote care (Shared House), science education (Glued Room), crew training (Station Zero).

Procedural generation?

Live: Roll a Lung completes a real airway with a published model and lays it out in H^2 in seconds. In general: A logical graph (cyclic grammar) plus a physical tile layout solved per visit so every door has free floor behind it; a modular kit and a style cascade. In data blades the data *is* the graph (airway tree, tree of life).

Is 4D useful?

Yes, where it has a job: time as the fourth axis (the Breath Room: the tumour as a tube through time whose shadow is the ITV), the w-coordinate behind overlapping rooms, and a 4D toybox. Not as the core renderer.

iPhone or Android?

iPhone: the 15 Pro Max as the headset phone and the 14 Pro Max as phone B (both have LiDAR). The Galaxy S24 as the Android partner. The challenge document doesn't require Huawei hardware.

Euclidean → non-Euclidean mapping?

Topology first: keep rooms flat and glue them with portals, wall gluings and sheets. For curved wings, place scanned meshes with the exponential map (or the Klein map where walls must stay flat), keeping objects ≤0.3R. The scanned room is the base chart; real doors become portals.

Pathfinding and placement?

Hierarchical A* on the room graph with transforms on the edges and graph-based heuristics (straight-line distance lies when portals exist); greedy routing in hyperbolic embeddings; a Pareto front for airway routes; clearance search on the sphere; objects stored in room-local coordinates.

Polished and coherent?

One learnable rule per blade; a stylised look with per-region colour grading, outlines, geodesic fog and baked light; a trim-sheet modular kit; scans de-lit and re-lit; audio beds and earcons; caption chips naming each geometry.

Phone–PC pairing?

The phone renders locally. LAN on your own router: WebSocket control, UDP pose, HTTP assets. USB tunnels for development. No video streaming.

How we measure (built in, costs the judge <30 s)

Comfort
Fast Motion Sickness Scale (one spoken item, 0–20), asked before and after.

Pointing probe
"Point back to the trachea door": the arrow is compared with the hyperbolic, Euclidean and graph predictions.

Ease
Single Ease Question (1–7) for the route task.

Logs
Frame times, forks, wraps, sheet changes, technique codes → Knife Map.

Hand to the implementation agent

ATLAS_Implementation_Brief.md	Full detail: decisions, blade specs, maths, architecture, message formats, build plan, formula sheet, key links. **Start with §1, §1.4 and §10.**
ATLAS_Research_Dossier.md	All 16 research notes, ≈850 links: purpose, rendering and 4D maths, locomotion, phone platform, networking, scanning, medical imaging, therapy, shared worlds, interaction design space
ATLAS_Engineering_Spec.md + .pdf	How to build it: stack, AR/VR pipeline, interfaces, data pipelines, tests, Phase 0 protocols, free Xcode signing
Challenge brief	Huawei Custom Challenge #1, "Beyond Euclid" (docx): rubric and bonuses

Key references

Non-Euclidean VR (Hart, Hawksley, Matsumoto, Segerman): arXiv 1702.04004, 1702.04862 · Holonomy, infinite H^2 walking in 3×3 m: arXiv 2409.05460 · HyperRogue tessellation engine: arXiv 2111.12040 · Warren 2017, wormhole navigation: PDF · Widdowson & Wang 2022, curved-space homing: PDF · Dolina 2008, bronchoscopic path selection: record · AAPM TG-76, respiratory motion: PDF · TCIA 4D-Lung: collection · TotalSegmentator: GitHub · Lui et al., Teichmüller maps: arXiv 1211.2569 · Choi 2024, telepresence placement: arXiv 2408.04297 · Cardboard SDK: GitHub · Apple RoomPlan: docs · Apple TN3179, local network privacy: docs · Ledo et al. 2018, evaluating HCI toolkits: PDF